

Will develop this project, connect to your local db

Complete Generative AI Course With Langchain and Huggingface

LangChain Chat with SQL DB

Choose suitable option

☒ Use SQLite3 Database-Student.db

☐ Connect to your SQL database

Groq API Key

Clear message history

How can I help you?

display all the records in the student table

✓ sql_db_list_tables: (empty string)

Thinking...

Action: sql_db_list_tables

Action Input: (empty string)

Ask me anything!

Deploy

✓ sql_db_query: SELECT * FROM student

Thought: I now have a query that I'm confident is correct, so I can execute it to get the desired result.

Action: sql_db_query

Action Input: SELECT * FROM student

[('Krish', 'Data Science', 'A', 90), ('John', 'Data Science', 'B', 100), ('Mukesh', 'Data Science', 'A', 86), ('Jacob', 'DEVOPS', 'A', 50), ('Dipesh', 'DEVOPS', 'A', 35)]

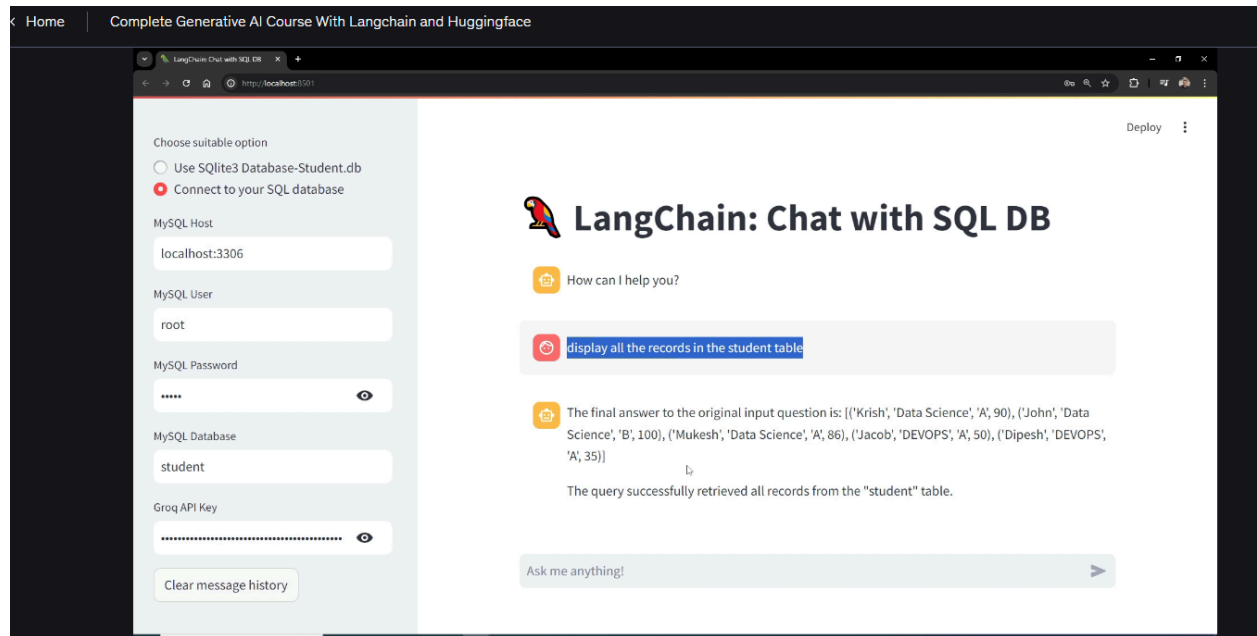
✓ Complete!

The final answer to the original input question is: [('Krish', 'Data Science', 'A', 90), ('John', 'Data Science', 'B', 100), ('Mukesh', 'Data Science', 'A', 86), ('Jacob', 'DEVOPS', 'A', 50), ('Dipesh', 'DEVOPS', 'A', 35)]

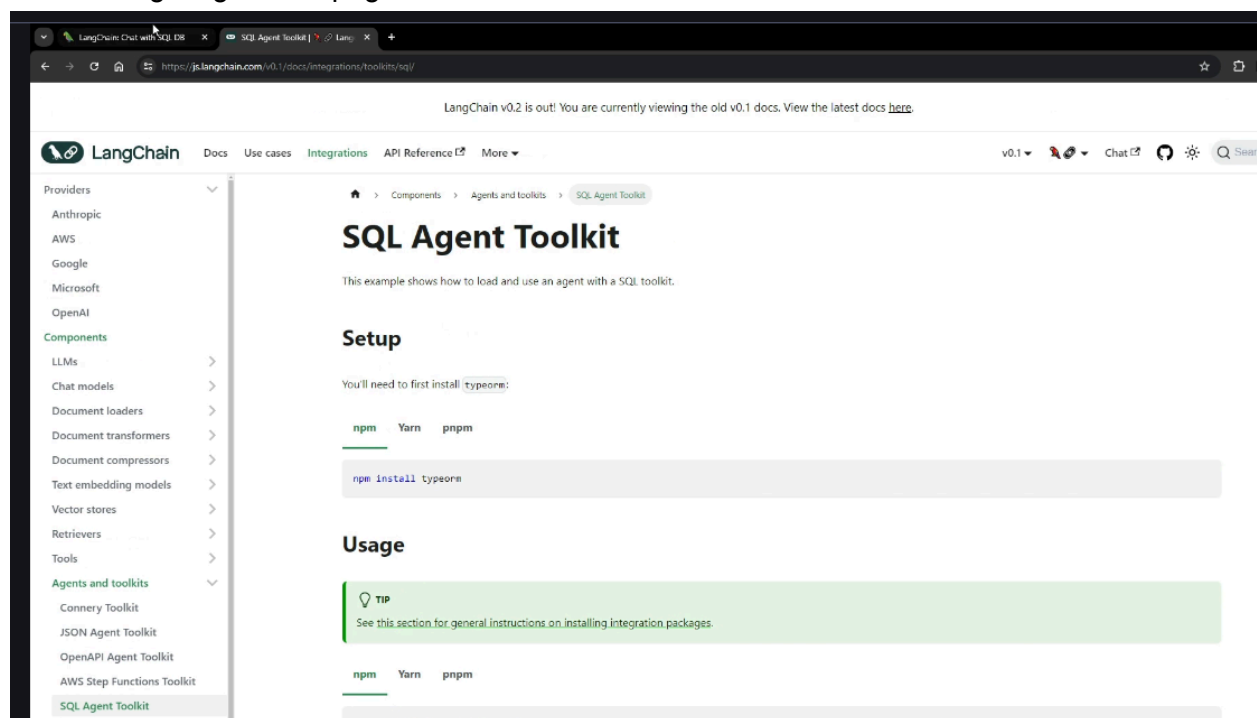
The query successfully retrieved all records from the "student" table.

Ask me anything!

connect to any other DB



will be using langchain sqlAgent toolkit



The screenshot shows the LangChain documentation website. The browser tabs include 'LangChain Chat with SQL DB', 'SQL Agent Toolkit', 'MySQL - Begin Your Download', 'Agents', 'LangChain', and 'langchain_community.agent_toolkits.sql.base.create_sql_agent.html'. The URL bar shows 'https://api.python.langchain.com/en/latest/agent_toolkits/langchain_community.agent_toolkits.sql.base.create_sql_agent.html'. The page has a navigation bar with 'LangChain', 'Core', 'Community', 'Experimental', 'Text splitters', 'Partner libs', and 'Docs'. A search bar is on the right. The main content area is titled 'langchain_community.agent_toolkits.sql.base.create_sql_agent' and shows the function signature: `langchain_community.agent_toolkits.sql.base.create_sql_agent(llm: BaseLanguageModel, toolkit: Optional[SQLDatabaseToolkit] = None, agent_type: Optional[Union[AgentType, Literal['openai-tools', 'tool-calling']]] = None, callback_manager: Optional[BaseCallbackManager] = None, prefix: Optional[str] = None, suffix: Optional[str] = None, format_instructions: Optional[str] = None, input_variables: Optional[List[str]] = None, top_k: int = 10, max_iterations: Optional[int] = 15, max_execution_time: Optional[float] = None, early_stopping_method: str = 'force', verbose: bool = False, agent_executor_kwargs: Optional[Dict[str, Any]] = None, extra_tools: Sequence[BaseTool] = [], *, db: Optional[SQLDatabase] = None, prompt: Optional[BasePromptTemplate] = None, **kwargs: Any) -> AgentExecutor`. Below the signature, it says 'Construct a SQL agent from an LLM and toolkit or database.' and lists parameters:

- llm** (*BaseLanguageModel*) – Language model to use for the agent. If agent_type is "tool-calling" then llm is expected to support tool calling.
- toolkit** (*Optional[SQLDatabaseToolkit]*) – SQLDatabaseToolkit for the agent to use. Must provide exactly one of 'toolkit' or 'db'. Specify 'toolkit' if you want to use a different model for the agent and the toolkit.
- agent_type** (*Optional[Union[AgentType, Literal['openai-tools', 'tool-calling']]]*) – One of "tool-calling", "openai-tools", "openai-functions", or "zero-shot-react-description". Defaults to "zero-shot-react-description". "tool-calling" is recommended over the legacy "openai-tools" and "openai-functions" types.
- callback_manager** (*Optional[BaseCallbackManager]*) – DEPRECATED. Pass "callbacks" key into 'agent_executor_kwargs' instead to pass constructor callbacks to AgentExecutor.
- prefix** (*Optional[str]*) – Prompt prefix string. Default depends on agent type.
- suffix** (*Optional[str]*) – Prompt suffix string. Default depends on agent type.
- format_instructions** (*Optional[str]*) – Formatting instructions to pass to ZeroShotAgent.create_prompt() when 'agent_type' is "zero-shot-react-description". Otherwise ignored.
- input_variables** (*Optional[List[str]]*) – DEPRECATED.
- top_k** (*int*) – Number of rows to query for by default.
- max_iterations** (*Optional[int]*) – Passed to AgentExecutor init.
- max_execution_time** (*Optional[float]*) – Passed to AgentExecutor init.

 On the right, there is an advertisement for 'Monetize your audience' by EthicalAds. The Windows taskbar at the bottom shows the time as 20:45 on 20/05/2024.

code logic is self explanatory